6. Deep Learning model performance (iteration 2); discussion of what went wrong and how to fix it

● Discuss how your strategy changed from last week based on the results Examples: Did you move to a different model architecture because you were overfitting? Did you add data transformation steps to your preprocessing pipeline?

We took out 20% of the total households as testing data and decided to reserve it for testing our final model at the end. This way we prevent data leakage.

We attempted to cluster the 700+ households into 10 representative samples in order for the model to take into account all the households we have. The clustering method we tried is to use piecewise aggregation approximation (PAA) to reduce data dimension and then use K-means to find the centroids. After finding the centroid households, we look for 10 real households that resemble the centroids the most to be the representative samples.

● Put your preprocessed data through the revised deep learning model architecture.

We ran the LSTM model on the 10 found representative samples and got the test MSE = 0.007, which is slightly worse than training on 10 randomly chosen households. We suspect that the dimension reduction method and clustering methods could be adjusted to improve the performance.

● List the performance metrics and any useful diagnostic figures

Testing the exact same model architecture, Pytorch offers 3-5x the speed of Tensorflow.

● Describe where the model worked well and where it didn't work well

So far, there is no substantial improvement in the average MSE.

The simpler testing models seem to minimize MSE when designed with a relatively small number of epochs (~3 epochs).